

The empirical recurrence for OEIS A299836, recovered from its tail

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Abstract

OEIS A299836 records “Empirical recurrence of order 68” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 87$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 70$. Its last coefficient is $c_{68} = -402 \neq 0$, so no further tail satisfies anything shorter; any order-68 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299836 is “Number of $n \times 5$ 0..1 arrays with every element equal to 1, 2, 3, 4, 5, 6 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 236, 7388, 198202, 5544044, 156175241, 4390323230, 123389988671, ...

The entry states:

Empirical recurrence of order 68 (see link above)

As of the “Last modified” line on the live entry (Feb 20 2018 08:13:40, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 87$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 87$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 68: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 68.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 174$ exact terms from $n = 2$ on, it returns order 68 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	26	c_{18}	−2892270374	c_{35}	730556110897	c_{52}	4714131625
c_2	30	c_{19}	37769146425	c_{36}	168921240033	c_{53}	237750044
c_3	836	c_{20}	104938137271	c_{37}	−487914273338	c_{54}	−1324686340
c_4	278	c_{21}	41200858061	c_{38}	−953357427259	c_{55}	−949769364
c_5	−17006	c_{22}	−261864273874	c_{39}	−621737460759	c_{56}	−283616693
c_6	−42472	c_{23}	−676325831636	c_{40}	−65028518282	c_{57}	144518129
c_7	−305238	c_{24}	−525390758290	c_{41}	92985103754	c_{58}	173089266
c_8	437894	c_{25}	907859927381	c_{42}	48733499398	c_{59}	82615884
c_9	5465966	c_{26}	2384639297896	c_{43}	−16187251994	c_{60}	38739572
c_{10}	814007	c_{27}	1715358079598	c_{44}	−39939277502	c_{61}	13981504
c_{11}	−20681428	c_{28}	−1128197940812	c_{45}	1846192014	c_{62}	−1368594
c_{12}	−34532404	c_{29}	−3641698761985	c_{46}	10429329709	c_{63}	−3240294
c_{13}	−28351748	c_{30}	−3003608533294	c_{47}	−11432747341	c_{64}	−1517101
c_{14}	472221391	c_{31}	−56679796914	c_{48}	−2135945644	c_{65}	−623481
c_{15}	381995765	c_{32}	1838950149096	c_{49}	8984571268	c_{66}	−178990
c_{16}	−2877008561	c_{33}	1975874915390	c_{50}	7642002538	c_{67}	−21695
c_{17}	−6548270966	c_{34}	1402817354204	c_{51}	7211670184	c_{68}	−402

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 70$, and it is the recurrence OEIS A299836 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{68} - c_1 x^{67} - \dots - c_{68}$. Its constant term is $-c_{68} = 402$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 68 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 68, so $\deg q = 68$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 68, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 68, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -402 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-68 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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